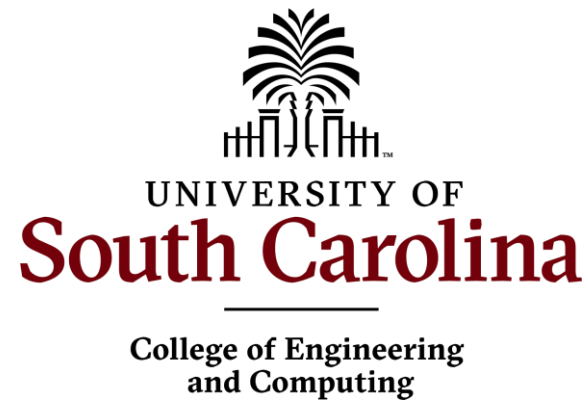


INTRODUCTION TO APPLIED NUMERICAL METHODS

Numerical Integration: Simpson's 3/8 Rule

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SIMPSON'S 3/8 RULE

The Simpson's 1/3 Rule:

- Only applicable to an even number of segments and an odd number of points

The Simpson's 3/8 Rule:

- Applicable to an odd number of segments and an even number of points

The Combination of Simpson's 1/3 and 3/8 Rule will allow evaluation of both cases

The Simpson's 3/8 Rule:

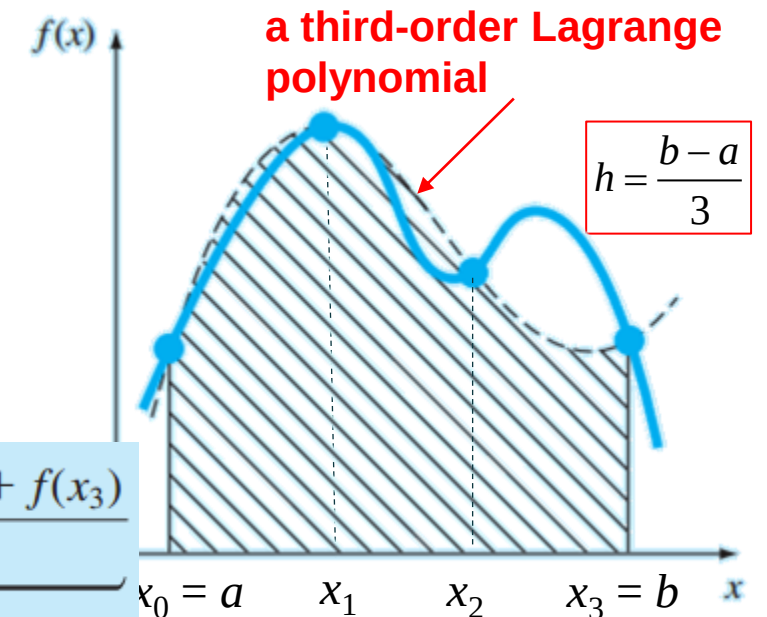
$$I = \int_a^b f(x) dx \cong \int_a^b f_3(x) dx$$

$$I \cong \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

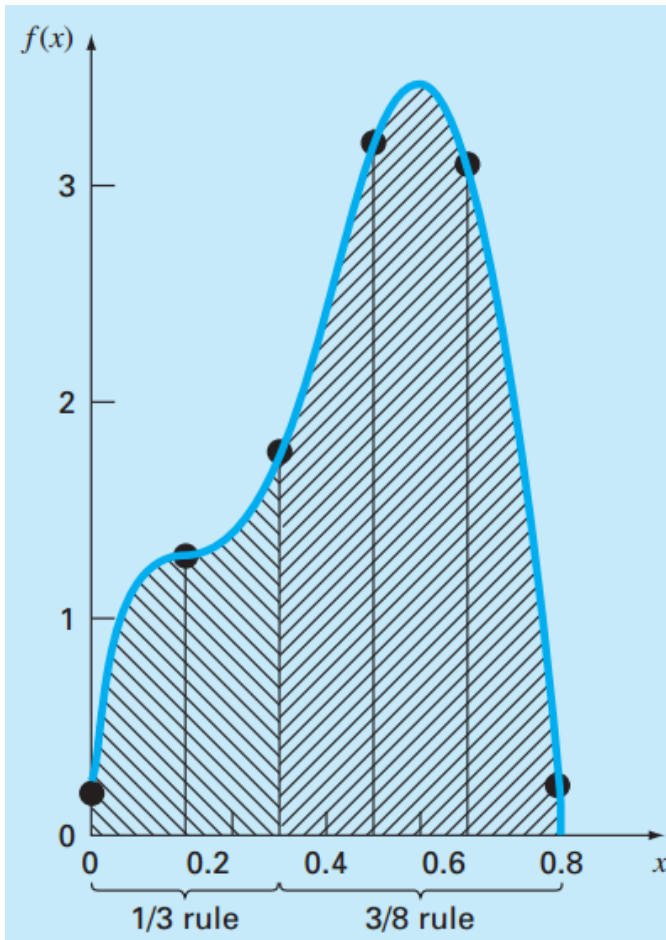
"3/8" rule
2

$$h = (b-a)/3$$

$$I \cong \underbrace{(b-a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)}{8}}_{\text{Average height}}$$



COMBINED SIMPSON'S 1/3 & 3/8 RULE



Five Segments:

- Five applications of trapezoidal rules (1 segment each)
- 2 applications of Simpson's 1/3 rule (2 segments each) and one application of trapezoidal rule (1 segment)
- 1 application of Simpson's 1/3 rule (2 segments) and one application of Simpson's 3/8 rule (3 segments)
- In general Simpson's rule is preferred over the trapezoidal rule because of higher accuracy enabled by the interpolating polynomial

AN EXAMPLE

Using the Simpson's 3/8 rule to numerically integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

From $a = 0$ and $b = 0.8$ (the exact value of integral is 1.640533).

Solution:

A single application of Simpson's 3/8 rule requires four equally spaced points:

$$x_0 = 0, \quad f(0) = 0.2$$

$$x_1 = 0.2667, \quad f(0.2667) = 1.432724$$

$$x_2 = 0.5333, \quad f(0.5333) = 3.487177$$

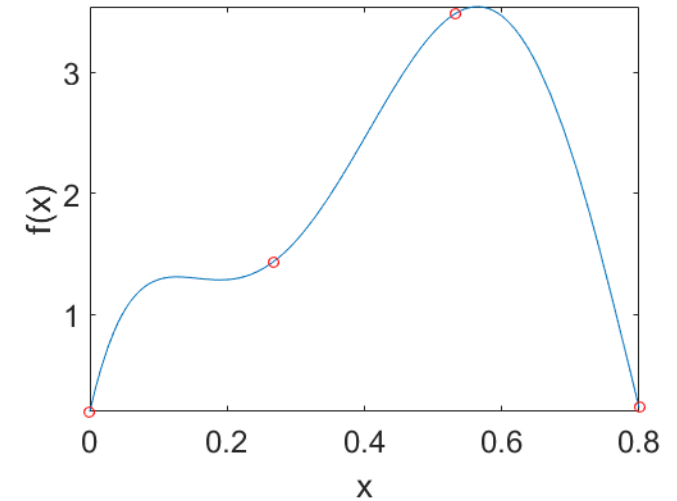
$$x_3 = 0.8, \quad f(0.8) = 0.232$$

$$I \cong \underbrace{(b - a)}_{\text{Width}} \underbrace{\frac{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)}{8}}_{\text{Average height}}$$

$$I \cong 0.8 \frac{0.2 + 3(1.432724 + 3.487177) + 0.232}{8} = 1.519170$$

$$E_t = 1.640533 - 1.519170 = 0.1213630$$

$$^4 \quad \varepsilon_t = 7.4\%$$

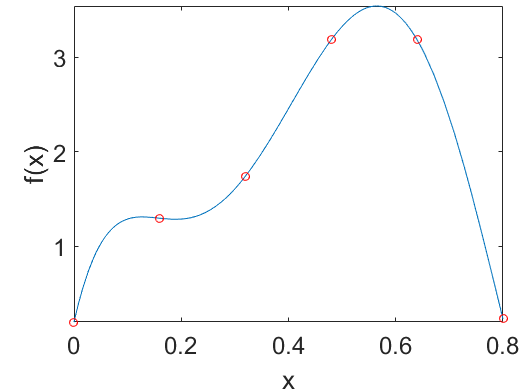


AN EXAMPLE

Using a combination of Simpson's 1/3 rule and 3/8 rule to numerically integrate

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

From $a = 0$ and $b = 0.8$ (the exact value of integral is 1.640533).



Solution:

The combined application requires six equally spaced points:

$$x_0 = 0, \quad f(0) = 0.2$$

$$x_1 = 0.16, \quad f(0.16) = 1.296919$$

$$x_2 = 0.32, \quad f(0.32) = 1.743393$$

$$x_3 = 0.48, \quad f(0.48) = 3.186015$$

$$x_4 = 0.64, \quad f(0.64) = 3.181929$$

$$x_5 = 0.8, \quad f(0.80) = 0.232$$

The 1st application of Simpson's 1/3 rule (3 pts):

$$I \cong 0.32 \frac{0.2 + 4(1.296919) + 1.743393}{6} = 0.3803237$$

The 2nd application of Simpson's 3/8 rule (4 pts):

$$I \cong 0.48 \frac{1.743393 + 3(3.186015 + 3.181929) + 0.232}{8} = 1.264754$$

The total integral is:

$$I = 0.3803237 + 1.264753 = 1.645077$$

$$E_t = 1.640533 - 1.645077 = -0.00454383 \quad \varepsilon_t = -0.28\%$$